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ANN based CMOS ASIC design for improved temperature-drift compensation of piezoresistive micro-machined high resolution pressure sensor

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ABSTRACT

The paper investigates the temperature-drift compensation of a high resolution piezoresistive pressure sensor using ANN based on conventional neuron model as also the inverse delayed function model of neuron. Using the delayed neuron model, an improvement in temperature-drift compensation has been obtained compared to the conventional neuron model. The CMOS analog ASIC design of a feed forward neural network using the inverse delayed function model of self connectionless neuron for the precise temperature-drift compensation has been presented. The inverse tan-sigmoid function is realized in CMOS implementation by Gilbert multiplier, differential adder and a cubing circuit. The entire design of the circuit has been done using AMS 0.35 μm CMOS model and simulated using Mentor Graphics ELDO simulator. Using the inverse delayed function model of neuron a mean square error of the order of 10^{-7} of the neural network has been obtained against a mean square error of the order of 10^{-3} using conventional neuron model for the same architecture of ANN. This brings down the error from 9% for uncompensated sensor to 0.1% only for compensated sensor using the delayed model of neuron in the temperature range of 0–70 °C. Using conventional neuron based ANN compensation, the error is reduced to 1% error.

1. Introduction

Silicon piezoresistive MEMS sensors are widely used for different applications like automobile, process control and biomedical systems mainly because of small size and low cost [1]. But silicon pressure sensor exhibits a non-linear dependence on temperature, an initial offset and strong drift of offset with temperature [2–4]. This non-ideal behavior of silicon piezoresistive MEMS sensor requires continuous monitoring and on line correction. The thermostatic control of temperature compensation of silicon pressure sensor has been reported [2]. The temperature-drift compensation of piezoresistive pressure sensor can be done using analog or digital signal conditioning circuits [3]. Digital signal conditioning circuits necessitate the use of A/D and D/A conversion blocks that leads to considerable consumption of chip area. On the other hand, analog compensation circuits can be implemented with input output polynomial relationship that is inverse of the sensor output function. The dependence of voltage output of piezoresistive sensor on the ambient pressure and temperature is given by [4],

$$V_{\text{out}} = P(\sigma_T T + \sigma_T^2 T^2) + (\text{Off} + \text{Off}_1 T + \text{Off}_2 T^2) \tag{1}$$

where P is the actual pressure and T is the actual temperature, σ_T is first order temperature coefficient, σ_T^2 is second order temperature coefficient, Off_0 is temperature independent offset, Off_1 is coefficient of first order temperature dependent offset, Off_2 is coefficient of second order temperature dependent offset. However these coefficients may have process variations from one batch to the other one as well as the entire range of pressure and temperature thus determination of pressure using Eq. (1) directly may lead to an erroneous value, particularly for the case of high resolution pressure sensors. An alternative approach is to employ an artificial neural network based signal conditioning block that can compensate for temperature-drift of output characteristics of a piezoresistive pressure sensor and offset over a chosen range of ambient temperature values and within a predetermined range of pressure values. ANN has therefore been used for temperature-drift, and offset compensation of piezoresistive pressure sensor. A review of contemporary literature also reveals the use of ANN for sensor signal conditioning with a limited accuracy of about ±1% [5–8]. However, for low pressure applications, precision signal conditioning of sensor output over the range of its operation is essential for achieving high resolution.

The current work presents an ASIC based scheme to compensate for the temperature error of high resolution piezoresistive silicon pressure sensor. The scheme under consideration is based on a

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CMOS ASIC realization of a feed forward artificial neural network. Such a signal conditioning scheme demands for a high accuracy especially for high resolution low pressure sensor. Conventional neural network when utilized for the signal conditioning applications yields a mean square error of about 10^{-3}. Also the error in temperature-drift ranges is minimized from 9% for uncompensated sensor to 1% in the temperature range of 0–70 °C. Neural network based compensation of sensor output characteristics to a similar extent has also been reported by Patra and Panda [7] and Islam et al. [8]. However, further improvement in accuracy is not possible with conventional neural network, in which the neurons have positive resistance effect that leads to reaching suboptimal solution of synaptic weights and still maintain a stability of the network. The inverse delayed model of neuron proposed by Hayakawa and Nakajima [9] is based on the model of a true biological neuron including the Bonhoeffer Vander Pol model [10]. The inverse delay model has a finite conversion time from the internal state of the unit to the output. The characteristic feature of inverse delayed function model is that the negative resistance effect is automatically incorporated in neuron model that can destabilize stable equilibrium points of the network within local optima leading towards the global optimal solution. The result is that a more accurate solution can be reached compared to the conventional neural network with limited number of neurons. The current network under consideration uses a simpler type self connectionless neuron unlike the self-connected type based on Hopfield model [9] and still achieves the negative resistance effect of inverse delayed function model. Self-connectionless neuron model is used for taking into account the hysteresis effect in the context of auto associative memory network [7]. In the case of temperature-drift compensation however there is no hysteresis effect involved. Hence self connectionless neuron model is selected in this work. The proposed analog CMOS circuit based on inverse delayed function model of self connectionless neuron incorporates intelligence to MEMS based sensor and transforms it into smart sensor. Using, the delayed neural network, the error in temperature compensation is further minimized from 1% to 0.1% in the temperature range of 0–70 °C.

The ANN can be implemented using digital or analog circuits. In the present work, analog circuit has been chosen for high speed, small chip area and low power consumption [11]. The neuron in the proposed neural network structure uses inverse tan-sigmoid activation function. The proposed network consists of two neurons in the input layer, five neurons in the hidden layer and one neuron in the output layer. The number of neuron in the hidden layer has been chosen heuristically. The selection of neuron is done on the basis of minimum mean square error, weight factor and bias values suitable for hardware implementation and its dynamic range. The network has been trained using the back propagation algorithm. Simulation studies are carried out in MATLAB and optimum synaptic weights have been found out.

The analog hardware implementation of ANN is done using inverse delayed function model of neuron. The multipliers required to implement multiplication of inputs with corresponding synaptic weights have been designed using Gilbert cell [12], whose gain can be controlled by application of control voltage. The choice of Gilbert cell as a multiplier is guided by the fact that a wide range of multiplication factor needs to be incorporated with minimum number of transistors. The summation of weighted inputs in each neuron has been implemented using compact high frequency differential adder proposed [13]. The inverse tan-sigmoid activation function have been designed using Gilbert cell [12], cubing circuit proposed by Shearer and MacEachern [14] and compact high frequency adder proposed [13]. The paper is organized into following broad sections. Section 2 focuses on the temperature dependence characteristic of piezoresistive pressure sensor, Section 3 describes the conventional neuron based ANN model for temperature-drift compensation of piezoresistive pressure sensor, Section 4 focuses on delayed neuron based ANN that has been proposed for further optimizing the temperature-drift compensation of pressure sensor. CMOS hardware implementation of neural network has been discussed in Section 5, and in Section 6 the simulation results has been presented.

2. Pressure sensor characteristics

In piezoresistive pressure sensors, the piezoresistors are placed in Wheatstone bridge configuration. Eq. (1) indicates dependence of output voltage on the ambient pressure and temperature of a piezoresistive pressure sensor bridge circuit. For matching the experimental characteristics reported [4] the different coefficients are chosen as $\sigma_T = 0.35 \times 10^{-3}$, $\sigma_T{}^2 = 0.017 \times 10^{-3}$, $\mathrm{Off} = \mathrm{Off}_1 = 1 \times 10^{-6}$ $\mathrm{Off}_2 = 0.013 \times 10^{-6}$. Using these coefficients, we get the output voltage vs. pressure characteristics for different temperatures as shown in Fig. 1.

This output voltage has a non-linear dependence on temperature. The non-linearity originates because of two factors. Firstly, the temperature coefficient of offset which primarily originates from resistor mismatch and residual stress causes a non-linear dependence of output voltage on temperature. Secondly, the temperature coefficient of sensitivity also causes non-linear dependence of output voltage on temperature [2,4]. These non idealities can be expressed in terms of error in uncompensated sensor output, both as a full scale output error (FSO) and as an offset error. These two errors for 35 °C reference are shown in percentage error vs. temperature plot in Fig. 2.

The temperature has been varied from 0 °C to 70 °C and offset error found to vary from 0% to 5.5% while the full scale output error is found to very from 0% to 3.5%. These errors can be compensated using analog or digital circuits. Digital compensation circuits necessitate the use of bulky A/D conversion blocks and look up tables that lead to considerable consumption of chip area and memory. This may pose problem when circuit is integrated with sensor. Analog signal conditioning circuits can be implemented with input output polynomial relationship that is inverse of sensor output function. However an exact inverse of the input output polynomial may be difficult analytically for high resolution measurements for such applications, the artificial neural network (ANN) can be used

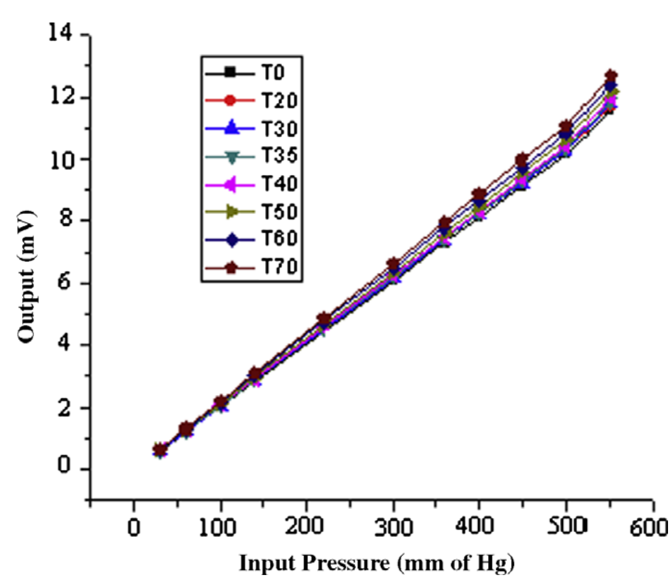

Fig. 1. Input pressure vs. output voltage of piezoresistive pressure sensor at different temperature.

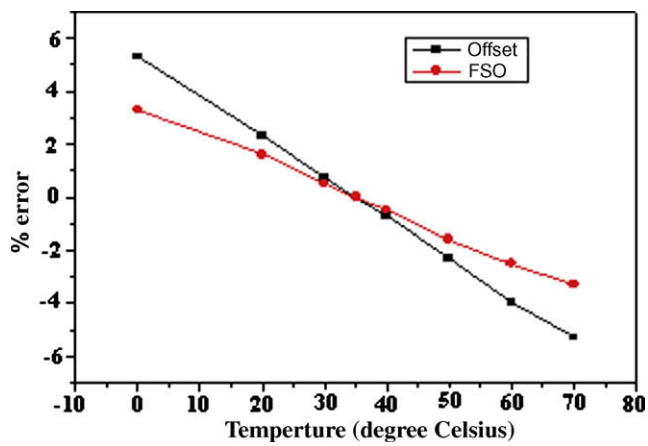

Fig. 2. Errors in uncompensated piezoresistive pressure sensor.

to compensate both the errors simultaneously. It can be implemented using digital or analog circuit. Digital ANN however has the same drawbacks as that of digital compensation. Hence the analog ANN has been chosen for high speed, small chip area and low power consumption [11].

3. ANN based temperature-drift compensation of pressure sensor using conventional neuron model

To compensate for the temperature effect on the output of sensor, the circuit should be modeled in such a way that its inverse characteristics, nullifies the effect of temperature on the output of sensor. This modeling can be done using inverse modeling of artificial neural network (ANN) [15]. Fig. 3 shows a block diagram of ANN based compensation scheme for piezoresistive pressure sensor using inverse model. In such model direct digital read out of the applied pressure is obtained by cascading the output of the piezoresistive pressure sensor to ANN in order to compensate for non-linearities in the output of characteristics due to temperature-drift [7]. For training the network, the standard back propagation learning algorithm can be used. Fig. 4 shows the MLP scheme of the ANN based on conventional neuron model The MLP ANN employs three layered perceptron based feed forward network with signal processing neurons in its hidden and output layers. The input layer has two inputs and bias, the number of neurons in hidden layer is five. The selection of the number of neurons in the hidden layer is done on the basis of minimum MSE, weight factor and bias values suitable for hardware implementation and its dynamic range. Moreover, the number of neurons has been kept as small as possible for simplicity in circuit implementation at the subsequent stage. The weight factors are chosen to be as small as possible for ease of design of Gilbert cell multiplier in subsequent ASIC design phase. The chosen training method is back propagation algorithm and the simulation studies are carried out in MATLAB, ANN tool box. The synaptic weights and bias values for proposed network are given in Table 1.

The voltage vs. pressure characteristics of the piezoresistive pressure sensor after compensation is shown in Fig. 5, corresponding to different temperature and pressure. From the error characteristics of Figs. 1 and 5 the % error in the compensated output response varies within 1% compared to about 9% error (combined) in the uncompensated sensor output for entire pressure and temperature range. The MSE is equal to 3.99×10^{-3}. The maximum % error corresponding to MSE in compensation at different temperature considering 35 °C as reference temperature is shown in Fig. 6.

Islam et al. and Patra and Panda have also reported neural network based compensation of sensor output characteristics to a similar extent [7,8]. Increasing the number of neurons minimizes the MSE only to a limited extent. However, further improvement in MSE is not possible using an ANN with conventional neurons [7]. This can be explained as follows. The error function (absolute error and not mean square error) of the neural network which leads to adjustments of synaptic weights is found to undulate with large number of varying sized crests and troughs at the initial stages of iterations in the training phase. Conventional neuron having positive resistance effect, after reaching suboptimal solution of synaptic weights and still maintains a stability of the network at relatively high value of ±1 in this case. So, in order to obtain optimal solution of synaptic weights, the neuron model is to be modified to destabilize the sub optimal local minima. This can be realized by incorporating finite conversion time from the internal state of the unit to the output. The characteristic feature of inverse delayed function model is that the negative resistance effect is automatically incorporated in the neuron model that can destabilize state equilibrium points of the network within local optima leading towards global optimum solution. The result is that a more accurate solution of synaptic weights can be obtained resulting into further minimization of error over the entire range.

4. ANN based temperature-drift compensation of pressure sensor using inverse delayed function model of neuron

As discussed in Section 3, the neural network is further modified to have inverse delayed function model of neurons [9]. However the neurons are assumed to be devoid of self connections unlike the one proposed by Hayakawa and Nakajima [9]. Hayakawa used self-connected neurons for the inverse delay model for taking into account of hysteresis effect in the context of auto associative memory network. However in the present work of temperature-drift compensation there is no hysteresis effect thus there is no need of self-connection of the neurons. The inverse delayed function model of neuron proposed by Hayakawa and Nakajima [9] is given by the following set of differential equations:

$$\tau \frac{du}{dt} = \sum_j w_{ij}x_j + a_{ii}x_i - u_i \tag{2}$$

$$\tau_x \frac{dx_i}{dt} = u_i - g(x_i) \tag{3}$$

$$g(x_i) = f^{-1}(x_i) - Kx_i \tag{4}$$

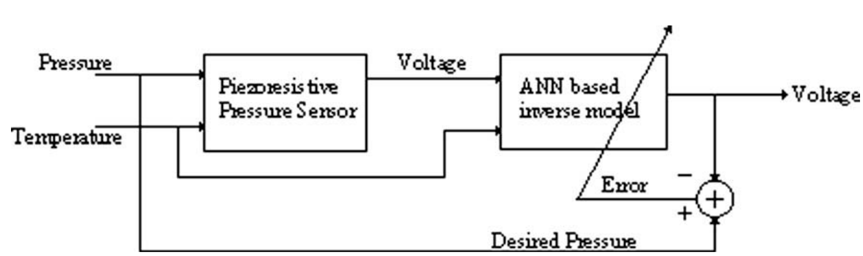

Fig. 3. Functional block diagram of ANN based compensation scheme for piezoresistive pressure sensor.

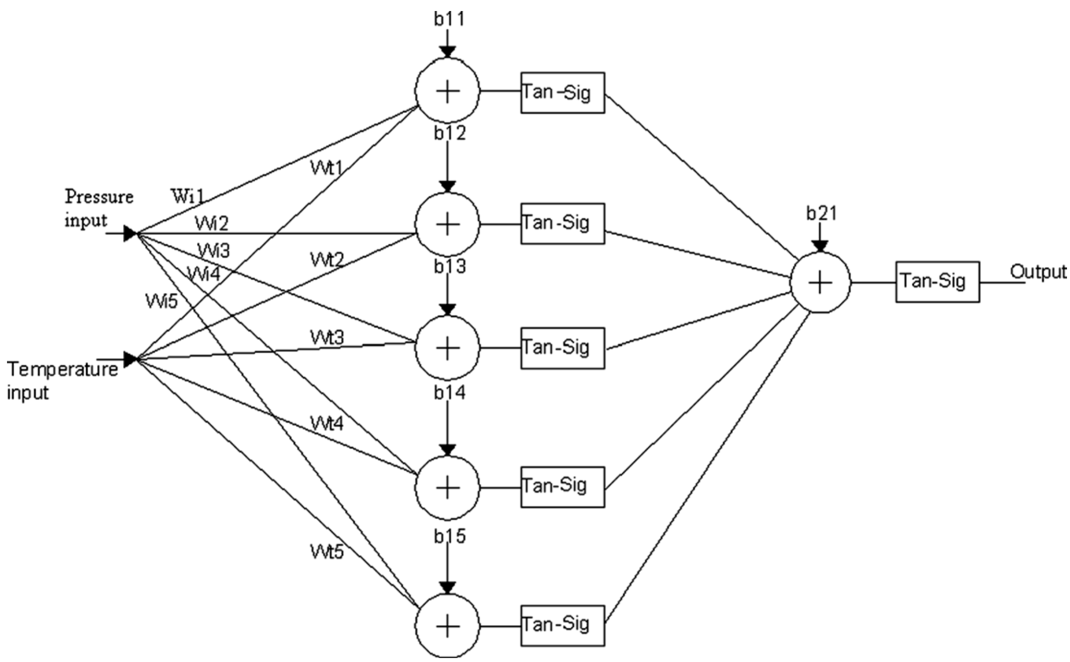

Fig. 4. MLA neural network using conventional neuron model.

Table 1
Synaptic weights and biases of the ANN based on conventional neuron model.

$iw\{1,1\}$ (input to hidden layer weights)	$iw^T\{2,1\}$ (hidden to output layer weights)	$b\{1\}$ (bias to hidden layer neurons)	$b\{2\}$ (bias to output layer neuron)
1.31 3.30	−3.23	−0.19	−0.07
3.00 −1.20	2.53	0.08	
−2.42 2.69	0.74	−0.35	
4.19 5.02	2.95	0.16	
2.21 3.00	−3.36	0.19	

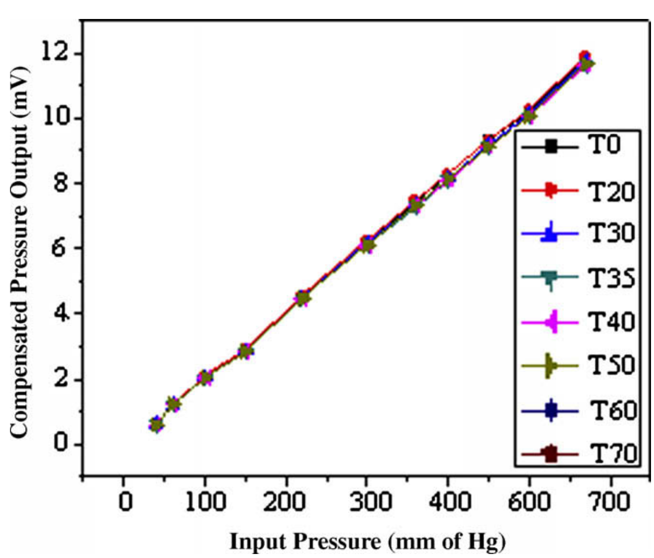

Fig. 5. Compensated output characteristics of pressure sensor using conventional neuron based ANN for different temperature.

where w_{ij} is the synaptic weight between ith and jth neurons, a_{ii} is the synaptic weight of self-connection, u_i is the internal state of ith neuron, x_j is the output of the jth neuron, τ and τ_x are the time constant of the internal state and time constant of the neuron output respectively. If $f(x)$ is the sigmoid function used in a conventional neural network, then $g(x) = f^{-1}(x)$ is the N-shaped inverse output function and $g(x)$ can be changed with the positive parameter K. The conversion from u to x has a very small response time compared to τ and thus $\tau_x \ll \tau$. However, the conversion time should take into consideration in general cases under the condition that it is much smaller than τ.

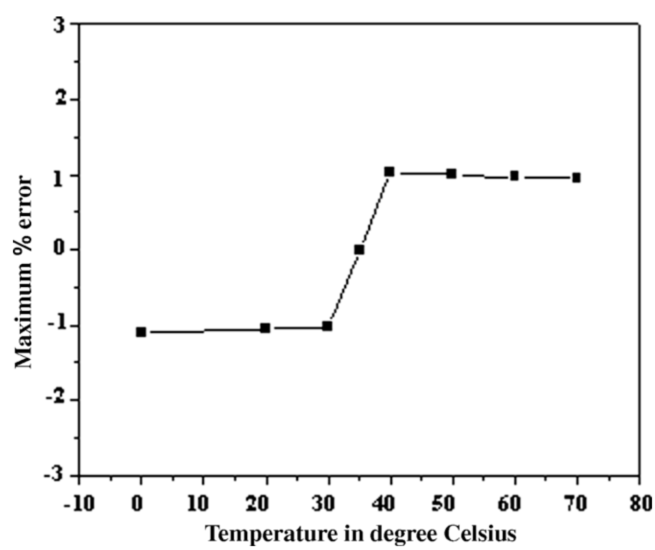

Fig. 6. Temperature vs. % error after compensation using conventional neuron model.

Since, the neurons in our network are devoid of self connections, we put $a_{ii} = 0$ in Eq. (2).

Differentiating Eq. (3) with respect to time t, we get,

$$\tau_x \frac{d^2 x_i}{dt^2} = \frac{du_i}{dt} - \frac{dg(x_i)}{dx_i} \cdot \frac{dx_i}{dt} \tag{5}$$

$$\text{Let,} \quad \phi_i = \frac{dg(x_i)}{dx_i} + \frac{\tau_x}{\tau} \tag{6}$$

Putting Eq. (5) in Eq. (4) we get,

$$\tau_x \frac{d^2x_i}{dt^2} + \phi_i \frac{dx_i}{dt} - \frac{\tau_x}{\tau} \cdot \frac{dx_i}{dt} = \frac{du_i}{dt}$$

$$\Longleftrightarrow \tau_x \frac{d^2x_i}{dt^2} + \phi_i \frac{dx_i}{dt} = \frac{1}{\tau}\left(\sum_j w_{ij}x_j - g(x_i)\right) \tag{7}$$

Let, $\dfrac{\partial U_i}{\partial x_i} = \dfrac{1}{\tau}\left(g(x_i) - \sum_j w_{ij}x_j\right)$

Therefore, Eq. (6) can be expressed as

$$\tau_x \frac{d^2x_i}{dt^2} + \phi_i \frac{dx_i}{dt} = -\frac{\partial U_i}{\partial x_i} \tag{8}$$

where $U_i = \frac{1}{\tau}(\int_0^{x_i} g(x_i)dx_i - x_i \sum_j w_{ij}x_j)$. U_i represents the potential of the inverse delayed function model. Eq. (8) corresponds to the kinetics of a particle in presence of frictional forces. The first term in Eq. (8) represents inertia and the second term represents friction. If $g(x_i)$ is an N-shaped function and $\frac{dg(x_i)}{dx_i}$ is less than $-\frac{\tau_x}{\tau}$ over a certain range of x_i, then the inverse delayed model has the effect of negative resistance ($\phi_i < 0$).

The energy function of the inverse delayed function model using Lyapunov function is given by,

$$E = -\frac{1}{2\tau}\sum_i \sum_j w_{ij}x_i x_j + \frac{1}{\tau}\sum_i \int_0^{x_i} g(x_i)dx_i + \frac{\tau_x}{2}\sum_i \left(\frac{dx_i}{dt}\right)^2 \tag{9}$$

where the self-connection between the neurons is ignored, since there is no self-connection between the neurons in our network. The last term appears in Eq. (9) because of time delay introduced in the inverse delayed function model. Differentiating both sides of Eq. (9) with respect to time t, we get,

$$\frac{dE}{dt} = -\sum_i \frac{dx_i}{dt}\left\{\frac{1}{\tau}\sum_j w_{ij}x_j - \frac{1}{\tau}g(x_i) - \tau_x \frac{d^2x_i}{dt^2}\right\}$$

$$\frac{dE}{dt} = -\sum_i \left(\frac{dg(x_i)}{dx_i} + \frac{\tau_x}{\tau}\right)\left(\frac{dx_i}{dt}\right)^2$$

$$\Longleftrightarrow -\sum_i \phi_i \left(\frac{dx_i}{dt}\right)^2 \tag{10}$$

From Eq. (10), we find that the time evolution of the system is a motion in the state space that seeks out minima in E and comes to a stop at all such points if ϕ_i is positive. However, if ϕ_i is negative, the energy function does not monotonically decrease and can climb up the potential hill because of the effect of negative resistance and can destabilize stable equilibrium points of the network within the negative resistance area. Hence, a neural network to answer an optimization problem can be constructed, as all units in the network should be at the corners of the solution space when its motion comes to a stop at the global minima. On the other hand, when it comes to a stop at the local minima, some of the units in the network are in the inside of the solution space. Therefore, it is expected to increase the possibility to get out from the local minima, if the network uses an inverse delayed function model of neuron.

The ANN structure of the inverse delayed neuron model having three layers as shown in Fig. 7.

The input layer has two inputs neurons, the number of neurons in hidden layer are five. The number of neurons in the hidden layer has been chosen heuristically. The selection of number of neurons in the hidden layer is done on the basis of least square error, weight factor and bias values suitable for hardware implementation and its dynamic range. The chosen training method is back propagation algorithm. Simulation studies are carried out in MATLAB programming. The MATLAB program is written for conventional neuron and for inverse delayed function model. The voltage vs. pressure characteristics of the piezoresistive pressure sensor after compensation using the delayed neural network is shown in Fig. 8, with different temperature as a parameters. The Inverse delayed model based ANN gives much better error performance than conventional neuron based ANN. The MSE of inverse delayed model based ANN is 3.99×10^{-7} compared to conventional neuron based ANN model whose MSE is 3.99×10^{-3} as discussed in Section 3. The weights and bias values for proposed network are given in Table 2.

Fig. 9 shows the comparison of % error in compensated at different temperature in the range of 0–70 °C for the proposed inverse

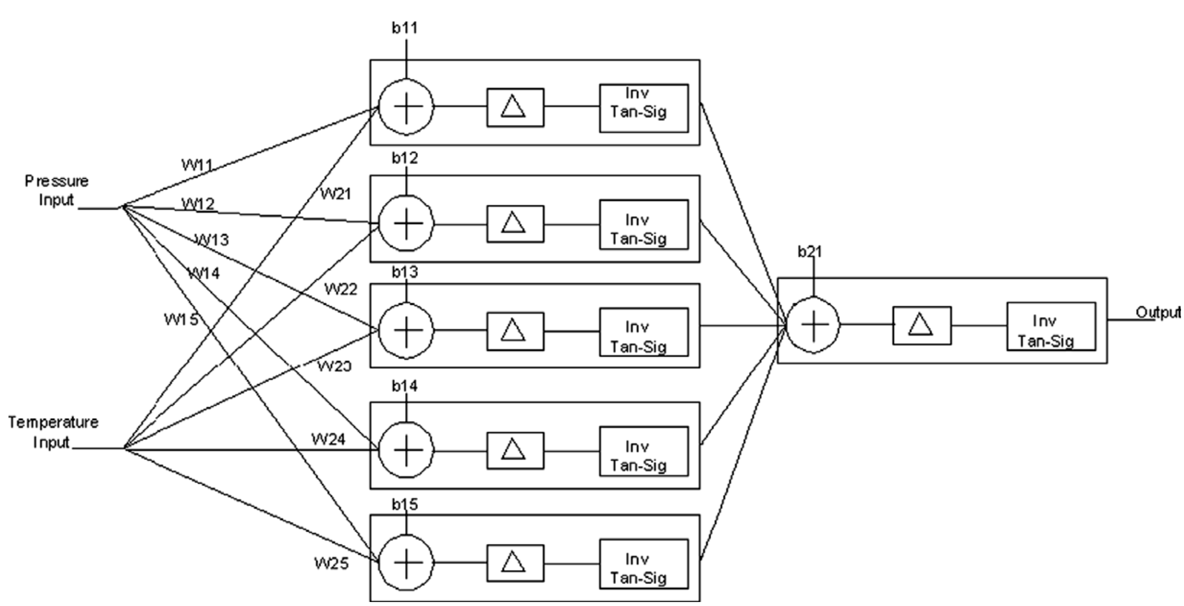

Fig. 7. MLP Neural Network using inverse delayed function model of neuron.

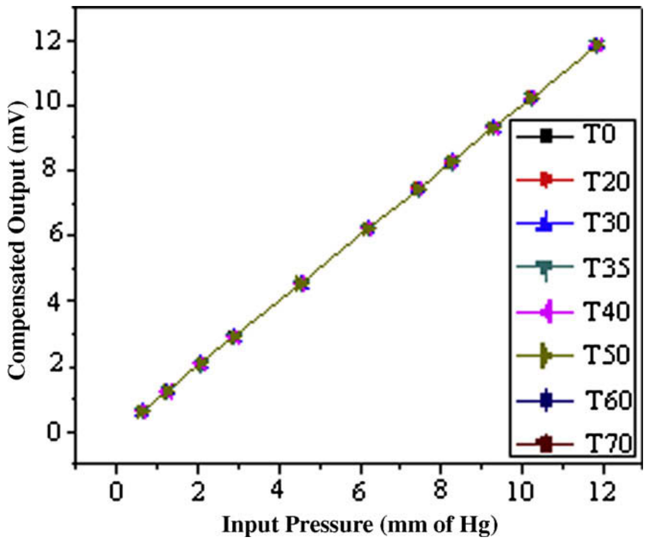

Fig. 8. Compensated output characteristics of pressure sensor using ANN based on ID model of neuron for different temperature.

Table 2
Synaptic weights and biases of the ANN based on delayed model of neuron.

iw{1,1} (input to hidden layer weights)	iw^T {2,1} (hidden to output layer weights)	*b*{1} (bias to hidden layer neurons)	*b*{2} (bias to output layer neuron)
1.20 3.30;	−1.01	−014	0.55
3.12 −1.19;	5.29	0.47	
−2.41 2.69;	2.61	−0.57	
4.34 5.10;	5.93	0.85	
2.09 2.99	−0.69	0.11	

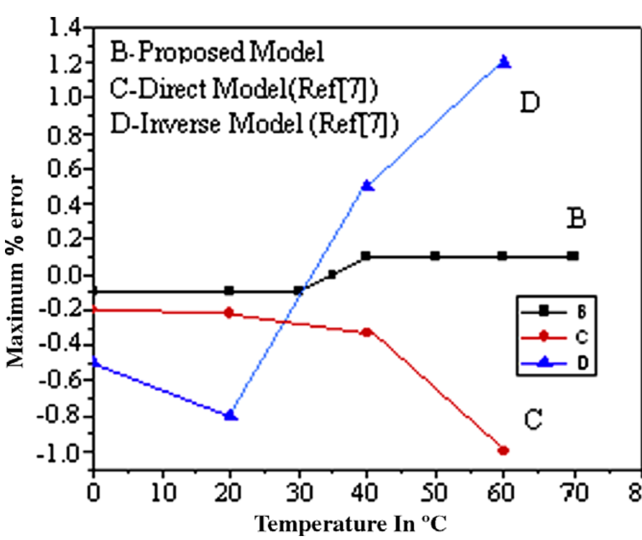

Fig. 9. Temperature vs. % error after compensation using (B) inverse delayed model of neuron, (C) direct and (D) inverse model of conventional model of neuron.

delayed model of neuron as well as the direct model and inverse model of conventional neuron as reported by Patra and Panda [7].

From the error characteristics we find that the error in the output response varies within ±0.1% using delayed neural network compared to about ±1% error in the compensated sensor output using ANN based on conventional neuron model as depicted in Fig. 6 and also ±1% error in compensated sensor output using the scheme presented [7]. The advantage of using the inverse delay model of neuron over the conventional neuron model can be explained as follows. The inverse delayed function model of neuron is to destabilize the local minimum states in the fitness function through its negative resistance effect which may prove to be a major issue in the optimization problems. Particularly, in ASIC design of neural network problems, while working with less number of neurons (for simplicity of ASIC design), the absolute error function of the neural network (which needs to be minimized) displays multiple local minima of varying depths. This may lead to a significant error in optimization if the local minima depths are significantly less than the global minimum. The use of inverse delayed function model destabilizes the stable equilibrium points at the local minima by introducing the negative resistance effect and stabilizes only at the global minimum. The negative resistance effect of inverse delayed function model of neurons has been incorporated by Hayakawa and Nakajima [9] through the use of inverse tan-sigmoid activation function.

The delayed neural network architecture is therefore chosen as a basis for the CMOS ASIC implementation for the compensation circuit.

5. Description of ASIC design of the proposed compensation scheme

In this section, the ASIC design of inverse delayed function model of neuron is described. The basic function of neuron is to amplify input signal proportionally to the weight value, to modify the *dc* signal value proportionally to the bias value, add these amplified signals and apply an inverse tan-sigmoid activation function to the amplitude of processed signal. Thus a neuron circuit should essentially consist of an analog multiplier an analog adder and a circuit to compute inverse tan-sigmoid activation function. A block diagram for the circuit of a neuron is shown in Fig. 10. It is to be noted that the synaptic weights of the neurons are determined off-line and the delay in the inverse delay neuron model is provided in programming for training purpose only. Delay in neuron model is provided for better optimization of synaptic weights and these optimized synaptic weights are directly used in multipliers. The neuron circuit therefore does not explicitly incorporate blocks to provide for the delay element. A survey of literature reveals some designs of analog multipliers reported [11,16]. The multiplier proposed by Hyogo et al. [16] employs as many as 32 transistors for four quadrant multiplication. Moreover, with a bias current of 12 μA, Hyogo et al. could achieve a multiplication factor up to 1. However, for our circuit implementation of neuron, a multiplication factor of up to 6 is needed. Therefore in order to achieve a linearity of multiplication over a wider range of multiplication factors, Gilbert cell multiplier has been chosen for our design. The circuit diagram of a single neuron is shown in Fig. 11. The

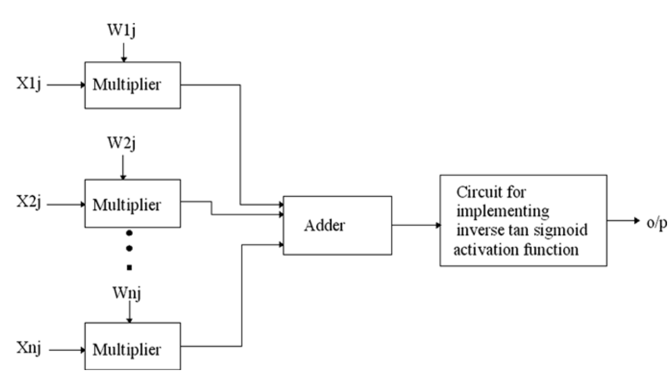

Fig. 10. Block diagram of neuron based on ID model.

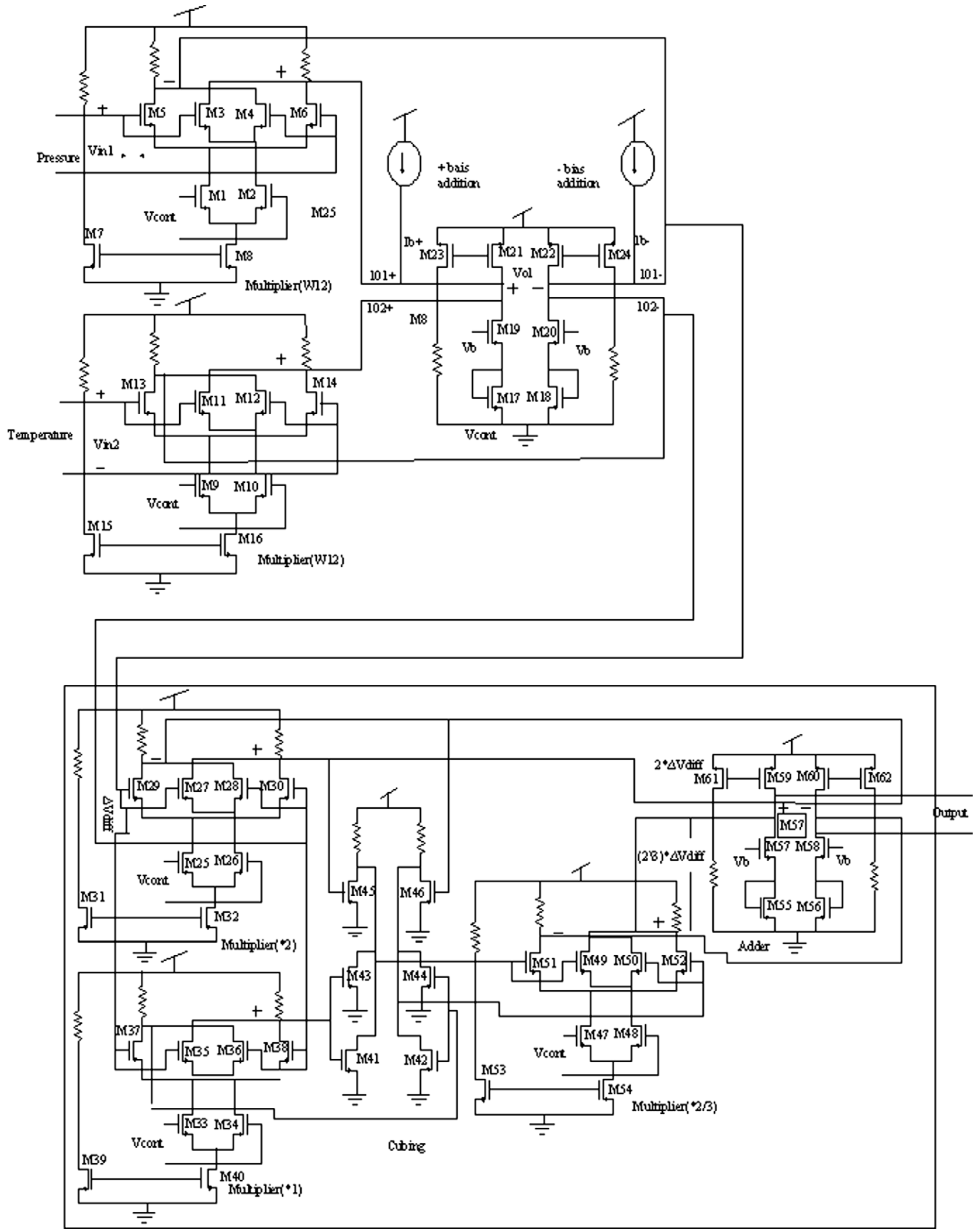

Fig. 11. CMOS implementation of single Neuron for simulation.

proposed neuron consists of Gilbert cell [12] as a multiplier whose output differential current is proportional to the input voltage V_{in} and multiplying factor can be controlled by V_{cont} The Gilbert cell is shown as a part of Fig. 11. The Gilbert cell has been chosen as multiplier for the proposed ASIC for its simplicity, more dynamic range and less power consumption and its ability to act as a four quadrant multiplier capable of handling the positive and negative weights. All transistors in Gilbert cell are biased in saturation region and obey the ideal square-law so that

$$\Delta I_{out} = KV^{*}_{in}V_{cont}; \qquad (11)$$

where K is a constant term dependent on the transistor size of differential pair. The typical values of W/L ratios of transistors in the Gilbert multiplier of Fig. 11 are

$$\left(\frac{W}{L}\right)_{M1} = \left(\frac{W}{L}\right)_{M2} = 134.4/1.4,$$
$$\left(\frac{W}{L}\right)_{M3} = \left(\frac{W}{L}\right)_{M4} = \left(\frac{W}{L}\right)_{M5} = \left(\frac{W}{L}\right)_{M6} = 67.2/1.4 \quad \text{and}$$
$$\left(\frac{W}{L}\right)_{M7} = \left(\frac{W}{L}\right)_{M8} = 268.4/1.4$$

These values have been computed from the chosen voltage and current biases of the different transistors of the Gilbert cell for 0.35 µm technology with 3.3 V supply voltage The Gilbert cell has been designed to multiply input signals ranging from −200 mV to 200 mV and multiplier factor in the range of ±10. The power dissipation of the multiplier is 1.4 mW using 0.35 µm technology. Although the power simulation results seem to be worse than that of Hyogo's multiplier [16], however, it is to be noted that the bias current of Hyogo's multiplier is 12 µA and is designed to support multiplication with multiplication factor less than 1. In our multiplier the multiplication factors needed are in the range of [−10, 10] and hence the bias current is much larger (50 µA). However, the power dissipation of the designed multiplier is 1.4 mW which is still much less than 9.5 mW as reported by Gatet et al. [11].

The differential adder used as same as that reported [13]. The differential adder is shown as a part of Fig. 11.

The adder circuit accepts the differential currents $\Delta I01$ and $\Delta I02$ produced by multipliers and current proportional to the bias offset ΔIb as inputs. The bias voltage Vb is 1 V. The transistor sizes of the differential adder are

$$\left(\frac{W}{L}\right)_{M17} = \left(\frac{W}{L}\right)_{M18} = 70/1.4,$$
$$\left(\frac{W}{L}\right)_{M19} = \left(\frac{W}{L}\right)_{M20} = \left(\frac{W}{L}\right)_{M21} = \left(\frac{W}{L}\right)_{M22} = 33/1.4 \quad \text{and}$$
$$\left(\frac{W}{L}\right)_{M23} = \left(\frac{W}{L}\right)_{M24} = 2.8/1.4$$

The two symmetrical output transistor acts as a current to voltage converter thus giving out as a voltage equal to ΔV_{diff} and given by

$$\Delta V_{\text{diff}} = K_2^*(\Delta I01 + \Delta I02 + \Delta Ib) \tag{12}$$

where K_2 is the proportionality coefficient introduced by adder cell, $\Delta I01 = (I01+) - (I01-)$, the output of Gilbert cell connected to pressure sensor, $\Delta I02 = (I02+) - (I02-)$, the output of Gilbert cell connected to temperature sensor and $\Delta Ib = (Ib+) - (Ib-)$ is the offset bias of NN.

Fig. 12 depicts simulation results of the single multiplier and adder cell for different multiplication factor and bias offset. Fig. 12 shows the range of voltages over which the adder and multiplier of the neuron shows a linear behavior in its characteristics. The adder and multiplier is found to operate linearly in the range of −40 mV to 40 mV.

Another important operator block of an ANN implementation is the non-linear activation function, which is normally a sigmoid type function for conventional neuron model. However, in case of inverse delay neuron model it is inverse tan-sigmoid function given by

$$Y = \log\left(\frac{1+x}{1-x}\right) = \log(1+x) - \log(1-x) \tag{13}$$

Expanding Eq. (13) we get

$$y = 2^*x + \left(\frac{2}{3}\right)^* x^3 + \left(\frac{2}{3}\right)^* x^5 + \tag{14}$$

In our case $x = \Delta V_{\text{diff}}$ and since it is typically of the order of 0–0.1 V, the higher order terms are ignored. Neglecting terms of order higher than 3 (which is a reasonable approximation, since x is very small), we get

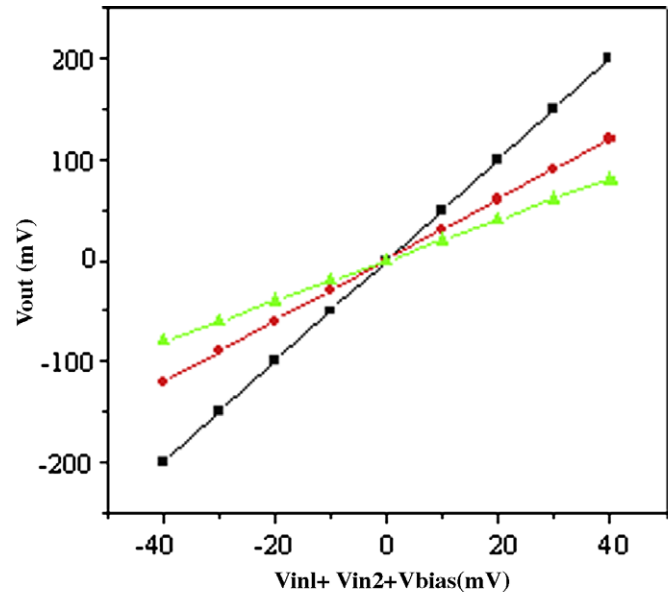

Fig. 12. Output response of multiplier and adder of neuron.

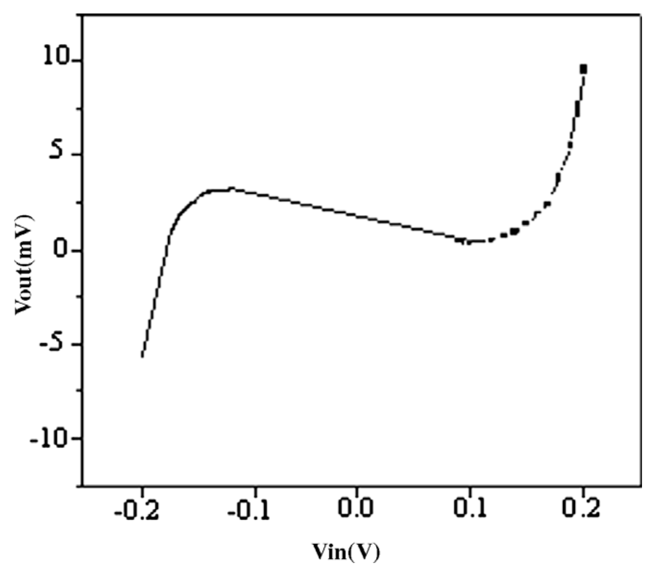

Fig. 13. Inverse tan-sigmoid function response.

$$y = 2^*x + (2/3)^*x^3 \tag{15}$$

Eq. (15) forms the basis of circuit implementation of inverse tan-sigmoidal activation function. The circuit accepts input from adder and produces output in terms of differential voltage given by

$$V_0 = 2^*\Delta V\text{diff} + (2/3)^*\Delta V\text{diff}^3$$

The implementation of this inverse tan-sigmoid function has been done with two multiplier using Gilbert cell and voltage cubing circuit [14]. This voltage cubing circuit offers significant reduction in total number of transistors required for cubing than the circuit used [16]. The complete inverse tan-sigmoid circuit is given as a part of Fig. 11.

In the implementation of cubing circuit, the inputs $\pm 2\,V_{\text{in}}$ and $\pm V_{\text{in}}$ are achieved using Gilbert cell. Each neuron consisting of multiplier, adder and activation function is tuned according to the weights and bias found from Matlab and are connected together to form a complete network using the same architecture as shown in Fig. 7. The (W/L) ratios of transistors are:

Table 3
Comparative study of ANN based compensation schemes.

ANN based compensation	Error (%)
ANN1 [7]	±1
ANN2 [8]	±1
ANN3 (our work based on conventional neuron model)	±1
ANN4 (our work based on ID neuron model)	±0.1

$$\left(\frac{W}{L}\right)_{M25} = \left(\frac{W}{L}\right)_{M26} = \left(\frac{W}{L}\right)_{M33} = \left(\frac{W}{L}\right)_{M34} = \left(\frac{W}{L}\right)_{M47} = \left(\frac{W}{L}\right)_{M48} = \frac{67.2}{1.4},$$

$$\left(\frac{W}{L}\right)_{M27} = \left(\frac{W}{L}\right)_{M28} = \left(\frac{W}{L}\right)_{M29} = \left(\frac{W}{L}\right)_{M30} = \left(\frac{W}{L}\right)_{M35} = \left(\frac{W}{L}\right)_{M36} = \left(\frac{W}{L}\right)_{M37} = \left(\frac{W}{L}\right)_{M38} = \left(\frac{W}{L}\right)_{M49} = \left(\frac{W}{L}\right)_{M50} = \left(\frac{W}{L}\right)_{M51} = \left(\frac{W}{L}\right)_{M52} = \frac{33.6}{1.4},$$

$$\left(\frac{W}{L}\right)_{M31} = \left(\frac{W}{L}\right)_{M32} = \left(\frac{W}{L}\right)_{M39} = \left(\frac{W}{L}\right)_{M40} = \left(\frac{W}{L}\right)_{M53} = \left(\frac{W}{L}\right)_{M54} = \frac{134.4}{1.4},$$

$$\left(\frac{W}{L}\right)_{M59} = \left(\frac{W}{L}\right)_{M60} = \left(\frac{W}{L}\right)_{M61} = \left(\frac{W}{L}\right)_{M62} = \frac{42}{1.4},$$

$$\left(\frac{W}{L}\right)_{M41} = \left(\frac{W}{L}\right)_{M42} = \left(\frac{W}{L}\right)_{M43} = \left(\frac{W}{L}\right)_{M44} = \left(\frac{W}{L}\right)_{M45} = \left(\frac{W}{L}\right)_{M46} = \left(\frac{W}{L}\right)_{M55} = \left(\frac{W}{L}\right)_{M56} = \left(\frac{W}{L}\right)_{M57} = \left(\frac{W}{L}\right)_{M58} = \frac{14}{1.4}$$

Fig. 13 shows the simulation output of the inverse tan-sigmoid module over the input range of ±200 mV with these optimized values of W/L.

6. Results and discussion

MATLAB simulation of ANN based on inverse delayed function model of neuron has been performed by MATLAB programming the Eqs. (2)–(4) with chosen values of $\tau = 0.52$ µs, $\tau_x = 0.025\tau$ and $K = 0$ corresponding to processor speed of 2.4 GHz Simulation result indicate that a mean square error of about 10^{-7} has been obtained with the inverse delayed model of neuron against a mean square error of about 10^{-3} obtained with an ANN based on the conventional neuron model. With this minimum value of mean square error, we get the optimal values of inter-neuronal synaptic weights and biases of the neural network using inverse delayed function model of neuron. Tables 1 and 2 show the weights and bias values of neural network based on both conventional and inverse delayed function model of neuron respectively. After selecting the weights and bias values, each neuron is designed to get desired weights with same common mode voltage level. The differential control voltages of Gilbert cells for corresponding multiplying factors are fixed with CMOS voltage reference maintaining common mode voltage level. The single neuron without activation function shown in Fig. 11, is simulated with inputs vin1 corresponding to pressure, vin2 corresponding to temperature and bias for different multiplying factors. The simulation results are shown in Figs. 12 and 13.

The circuit response shows proportional change in output with respect to input and corresponding to different multiplying factors. The simulation response of circuit implementing inverse tan-sigmoid activation function is shown in Fig. 13. The *N*-shaped characteristics of the inverse tan-sigmoid activation function is ideally suited to provide the negative resistance effect necessary in ID model of neuron.

The CMOS implementation of a single neuron with activation function of inverse tan-sigmoid function is simulated for the input voltage vin1 in the range of 0–13 mV, corresponding to pressure in the range of 0–600 mm of Hg and other input voltage vin2 in the range of 0–7 mV corresponding to temperature range of 0–70 °C. The power consumption of single neuron including inverse tan-sigmoid activation function block is 5.4 mW. All the neurons are tuned to corresponding weights, biases and are connected together to form a MLP structure corresponding to the ANN architecture as shown in Fig. 7 (see Table 3).

7. Conclusion

In this paper we have proposed an ANN based on inverse delayed function model of neuron for improved temperature-drift compensation for piezoresistive micro-machined high resolution silicon pressure sensor over the temperature range of 0–70 °C as compared to that obtained with conventional neuron based ANN. The observed error of proposed model is less than 0.1% over its full scale value compared to about 9% error for uncompensated sensor output and 1% error using ANN based on conventional neuron model. The analog CMOS ASIC implementation of the inverse delayed neuron model of neuron is reported for the first time. The Gilbert cell acts as the multiplier. Inverse tan-sigmoid function is realized by using two multipliers and one cubing circuit. As the proposed model is fully analog it eliminates the requirement of ADC and look up table thus increasing bandwidth. It also limits the chip size and power consumption over digital implementation. The total power consumption of the single neuron circuit is 5 mW. The proposed CMOS ASIC implementation is integrated with sensor to achieve desired temperature-drift compensation which can make sensor an excellent SMART piezoresistive MEMS pressure sensor.

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